FPGA BASED GAS LEAKAGE DETECTION AND ALERT SYSTEM

INTRODUCTION:

The CO_2 Level Indicator is a digital system designed using Verilog HDL to monitor carbon dioxide levels.

Increasing CO_2 concentration is an important environmental and indoor-air-quality concern associated with human activities and poor ventilation.

The system classifies the digital input into Safe, Warning, and Danger conditions.

LED indicators and a buzzer provide a simple alert when the CO_2 level reaches the danger range.

WORKING PROCESS:

- The system receives a 14-bit digital value representing the Carbon Dioxide level.
- The Verilog logic continuously compares the input value with predefined thresholds.
- 0–1000 is classified as the Safe level, and the green LED is turned ON.
- 1001–5000 is classified as Warning, and the yellow LED is turned ON.
- Above 5000 is classified as Danger, activating the red LED and buzzer.

CODING:

```
module CO2_led_level (
input  wire [13:0] value,
output reg green_led,
output reg yellow_led,
output reg red_led,
output reg buzzer);
always @(*) begin
   green_led  = 1'b0;
   yellow_led = 1'b0;
```

```
    red_led    = 1'b0;
    buzzer     = 1'b0;
    if ((value <14’d1000) begin
       green_led = 1'b1;
    end
    else if ((value >= 14'd1000)  && ( value<14’d5000))begin
       yellow_led = 1'b1;
    end
    else begin
       red_led = 1'b1;
       buzzer  = 1'b1;
    end
end
endmodule
```

FPGA CONNECTIONS:

